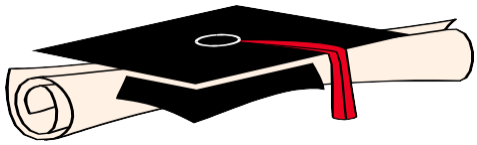
PCS

Programming with C

Semester: 2	**The Exam Code:** 01
Duration: 90 minutes	**Total Marks:** 20



Practical Examination Paper

Do not write on this question paper and return it to the Invigilator after the examination.

C# Programming Exam: Employee Management System

Total Points: 20

Instructions

- Ensure your code is well-structured, easy to read, and includes appropriate comments explaining your logic.
- Properly handle exceptions and edge cases to ensure the robustness of your program.
- Test your code to validate all functionalities.

Question 1: Create an Interface *(2 Points)*

Task: Create an interface `IEmployee` that includes the following methods:

1. `float CalculateBonus()`: Calculate the bonus of the employee.
2. `float CalculateTotalSalary()`: Calculate the total salary of the employee (`baseSalary + bonus`).
3. `void DisplayInfo()`: Display the details of the employee.

Question 2: Create the Employee Class *(6 Points)*

Task: Create a class `Employee` that implements the `IEmployee` interface.

Fields and Properties:

- Declare the following private fields:

Field	Type	Description
`id`	`int`	Unique ID of the employee
`name`	`string`	Name of the employee (Length: 1–50)
`position`	`string`	Employee's position (Length: 1–30)
`baseSalary`	`float`	The base salary (> 0)
`experienceYears`	`int`	Years of experience (>= 0)

[2 Points]: Validate constraints for each field using getters and setters:

- Ensure `name` and `position` lengths are valid.
- Ensure `baseSalary` is greater than 0.
- Ensure `experienceYears` is non-negative.

Methods:

- Implement the methods from `IEmployee`:
  - **`CalculateBonus()`** *(1.5 Points)*:
    Calculate bonus based on experience:
    - Less than 5 years: `baseSalary * 10%`.
    - 5–10 years: `baseSalary * 15%`.
    - Over 10 years: `baseSalary * 20%`.
  - **`CalculateTotalSalary()`** *(1 Point)*:
    Formula: `TotalSalary = baseSalary + bonus`.

`DisplayInfo()` *(1.5 Points)*:

Output example:
`Employee John works as Developer with 7 years of experience, earning a total salary of $7500.`

Question 3: Create the Manager Class *(4 Points)*

Task: Create a class `Manager` that inherits from `Employee`.

Fields and Properties:

- Additional Field:

Field	Type	Description
`teamSize`	`int`	The size of the team managed (>= 0).

Override Methods:

- **`CalculateBonus()`** *(2 Points)*:
  Add an extra bonus based on `teamSize`:
  - Small team (< 5): Add `$500`.

    - Medium team (5–10): Add `$1000`.
    - Large team (> 10): Add `$2000`.

`DisplayInfo()` *(2 Points)*:
Include `teamSize` in the output.
Example:

```
Manager Sarah leads a team of 8 members, earns a total salary of $8500.
```

Question 4: CRUD Operations on Employee *(5 Points)*

Task:
Implement the following CRUD functionalities in a new class `EmployeeManagement`:

- **Create Employee** *(1.5 Points)*:
    - Allow the user to add a new `Employee` or `Manager` to a list.
    - Input validation should follow the rules defined in the `Employee` class.
- **Read Employee Details** *(1.5 Points)*:
    - Display the list of all employees and their details.
    - Format the output for better readability.
- **Update Employee** *(1 Point)*:
    - Allow the user to update the details of an employee by ID.
    - Validate the updated values.
- **Delete Employee** *(1 Point)*:
    - Allow the user to remove an employee by ID.
    - Display an appropriate message if the ID is not found.

Question 5: Create a Test Class *(3 Points)*

Task:

- [1 Point]: Implement a menu-based program in a `Main` class to allow the user to perform CRUD operations.
- [1 Point]: Calculate and display the total salary of a selected employee or manager.
- [1 Point]: Catch and handle exceptions for invalid inputs (e.g., negative salary, duplicate ID).

Menu Example:

```
=== EMPLOYEE MANAGEMENT MENU ===

1. Add New Employee/Manager

2. View All Employees

3. Update Employee Details

4. Delete Employee

5. Calculate Total Salary

6. Exit

Your choice:
```

- Option 1: Execute the **Create Employee** functionality.
- Option 2: Execute the **Read Employee Details** functionality.
- Option 3: Execute the **Update Employee** functionality.
- Option 4: Execute the **Delete Employee** functionality.
- Option 5: Prompt the user to enter an employee ID and display the total salary.